ALICE L WALTON SCHOOL OF MEDICINE

Protein Synthesis Inhibitors–2 of 2: Adverse Drug Reaction

Last updated June 13, 2025 · 9 min read CC BY-SA 4.0

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Adapted from: Protein Synthesis Inhibitors

Listen to this Brick
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Learning Objectives (2) Versions

After completing this brick, you will be able to:

1. Recognize the common adverse reactions associated with protein synthesis inhibitors.
2. Identify the common drug interactions associated with protein synthesis inhibitors.

CASE CONNECTION

KH tells you, "I can deal with a little bit of acne," she says, with an embarrassed look toward her mother. "But I've reached the point where I don't really like to go out because of my face. I'm a senior, and I don't want to look like this at graduation." KH washes her face daily with ordinary soap. Her face has severe pustular acne. You decide KH would benefit from treatment with oral tetracycline. You explain to her that the tetracycline is an antibiotic that will fight the inflammation and reduce the bacteria that are causing most of the trouble.

What adverse effects will you discuss with KH? Consider your answer as you read, and we'll revisit KH at the end of the brick.

GO TO CONCLUSION ↓

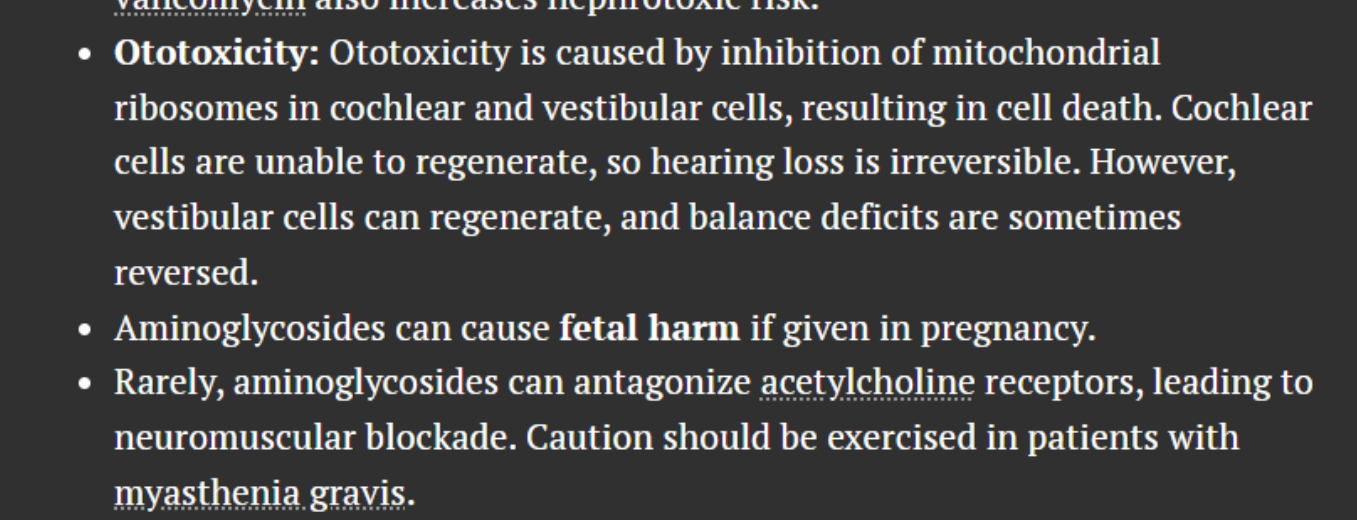
In the previous bricks on protein synthesis inhibitors, we discussed the mechanism of action of protein synthesis inhibitors (PSIs). In this brick, we are going to discuss the adverse effects that are associated with the use of these agents.

What Are Common Adverse Drug Reactions associated with protein synthesis inhibitors? (LO#1 and 2)

The ADRs of protein synthesis inhibitors is discussed by drug class.

Aminoglycosides

- Two common adverse reactions of aminoglycosides are **nephrotoxicity** and **ototoxicity**.
- **Nephrotoxicity:** Aminoglycosides are capable of filtering across the glomerulus and can accumulate in the proximal convoluted tubule (PCT). This accumulation increases the permeability of those cells, which can lead to acute tubular necrosis. The nephrotoxicity risk increases with impaired renal function and in the elderly. Concurrent use of other nephrotoxic drugs such as Amphotericin B, polymyxins, NSAIDS, radiocontrast dye, and vancomycin also increases nephrotoxic risk.
- **Ototoxicity:** Ototoxicity is caused by inhibition of mitochondrial ribosomes in cochlear and vestibular cells, resulting in cell death. Cochlear cells are unable to regenerate, so hearing loss is irreversible. However, vestibular cells can regenerate, and balance deficits are sometimes reversed.
- Aminoglycosides can cause **fetal harm** if given in pregnancy.
- Rarely, aminoglycosides can antagonize acetylcholine receptors, leading to neuromuscular blockade. Caution should be exercised in patients with myasthenia gravis.


Common Adverse Reactions of: Aminoglycosides
Nephrotoxicity
Ototoxicity
Fetal Harm

CLINICAL CORRELATION

Myasthenia Gravis (MG) is an autoimmune neuromuscular disease that classically presents as muscle weakness that worsens with usage. The disease inhibits the communication at the neuromuscular junction. Normally, acetylcholine (ACh) is released from a nerve and activates receptors on muscles to initiate contraction. However, in MG, antibodies alter the receptors and interrupt ACh binding (Figure 4). An aminoglycoside neuromuscular blockade would directly compound the existing deficit found in patients with MG.

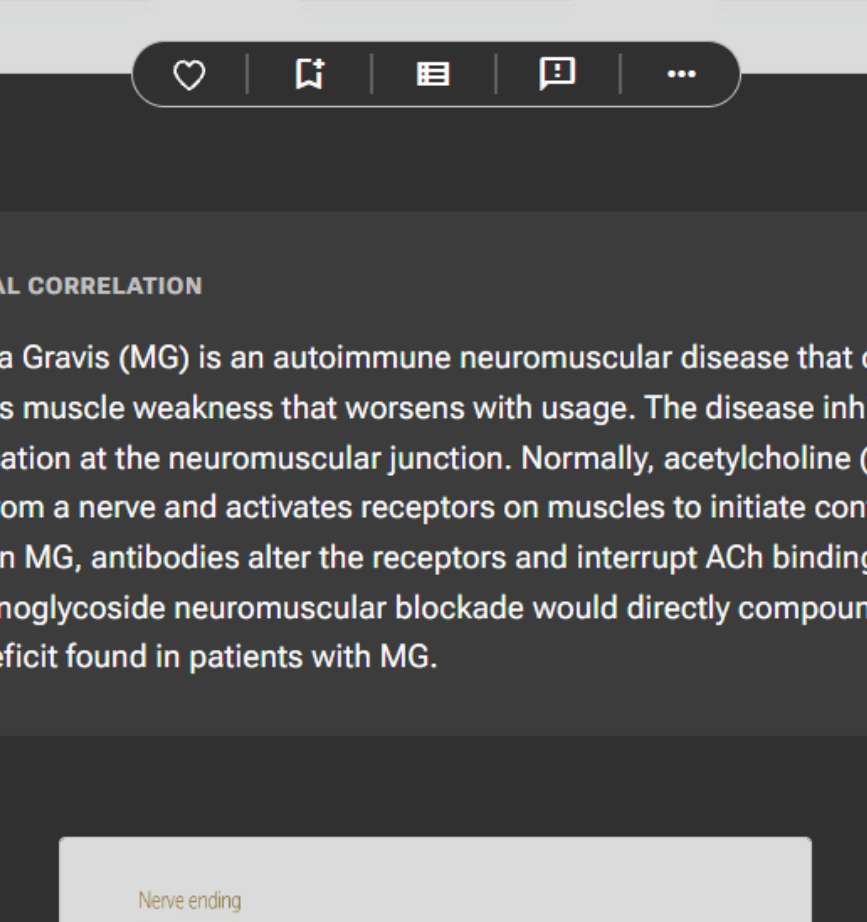
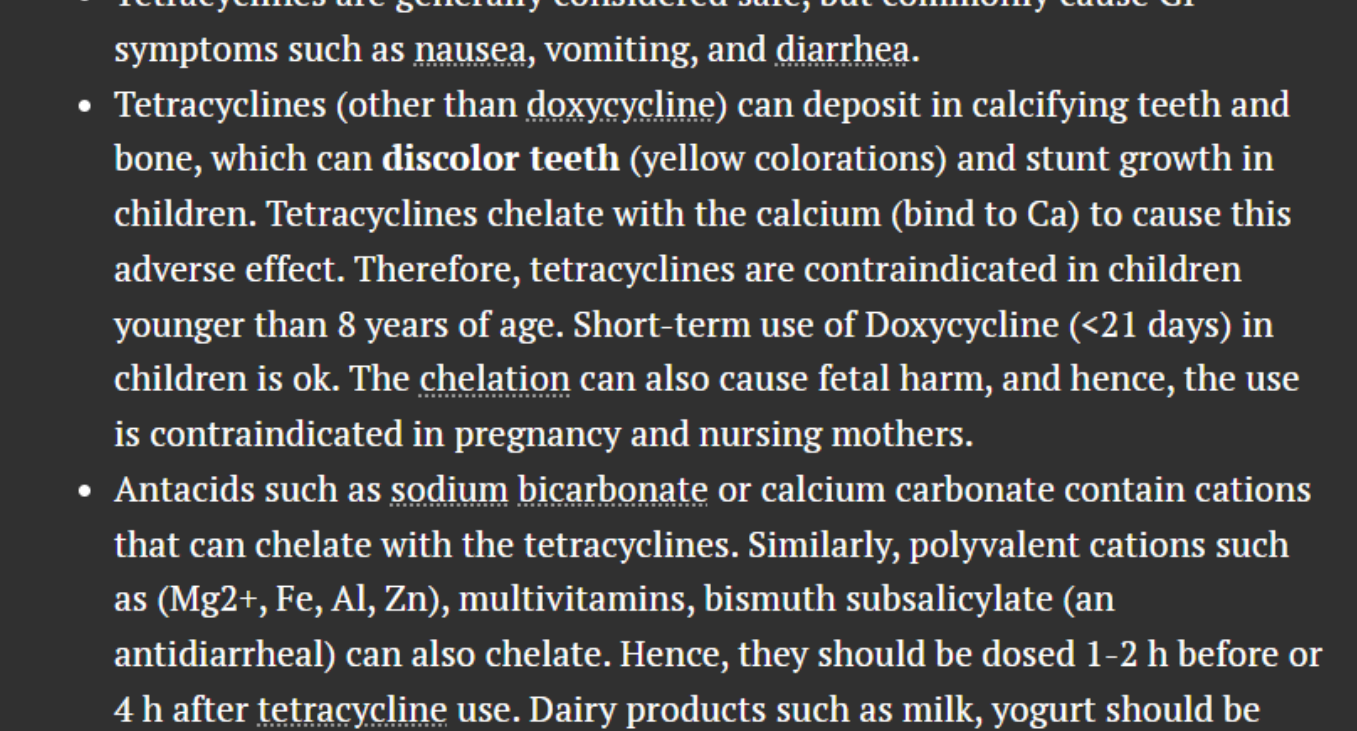


Figure 4

Tetracyclines

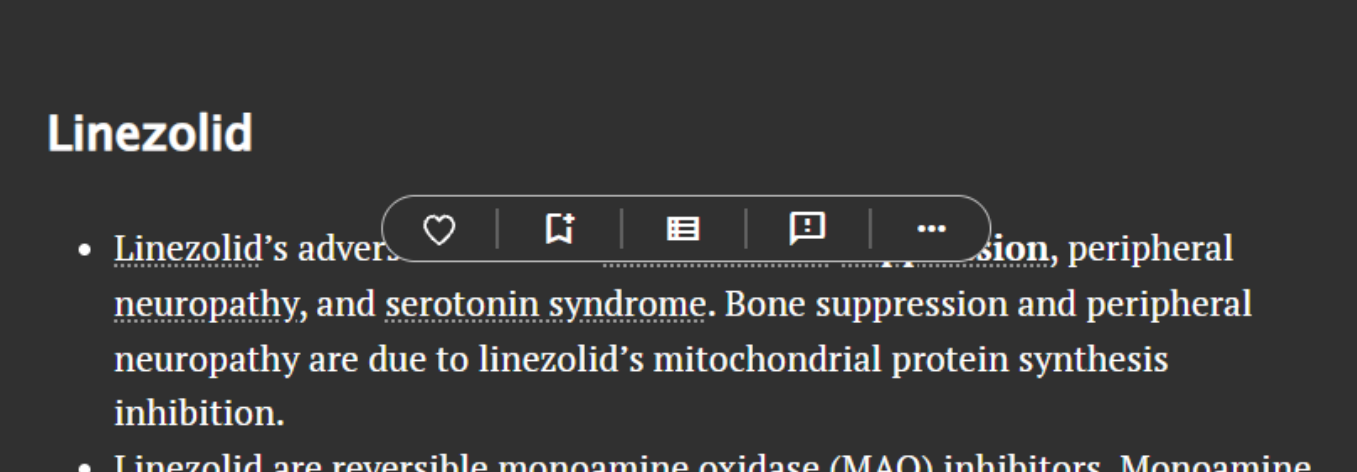
- Tetracyclines are generally considered safe, but commonly cause GI symptoms such as nausea, vomiting, and diarrhea.
- Tetracyclines (other than doxycycline) can deposit in calcifying teeth and bone, which can **discolor teeth** (yellow colorations) and stunt growth in children. Tetracyclines chelate with the calcium (bind to Ca) to cause this adverse effect. Therefore, tetracyclines are contraindicated in children younger than 8 years of age. Short-term use of Doxycycline (<21 days) in children is ok. The chelation can also cause fetal harm, and hence, the use is contraindicated in pregnancy and nursing mothers.
- Antacids such as sodium bicarbonate or calcium carbonate contain cations that can chelate with the tetracyclines. Similarly, polyvalent cations such as (Mg2+, Fe, Al, Zn), multivitamins, bismuth subsalicylate (an antidiarrheal) can also chelate. Hence, they should be dosed 1-2 h before or 4 h after tetracycline use. Dairy products such as milk, yogurt should be avoided 1 h before or 2 h after tetracycline.
- Tetracyclines are associated with **photosensitivity reactions** after sunlight (UV) exposure. Patients should be advised to avoid excessive sunlight



Why should tetracyclines be avoided in children under 8 years old?

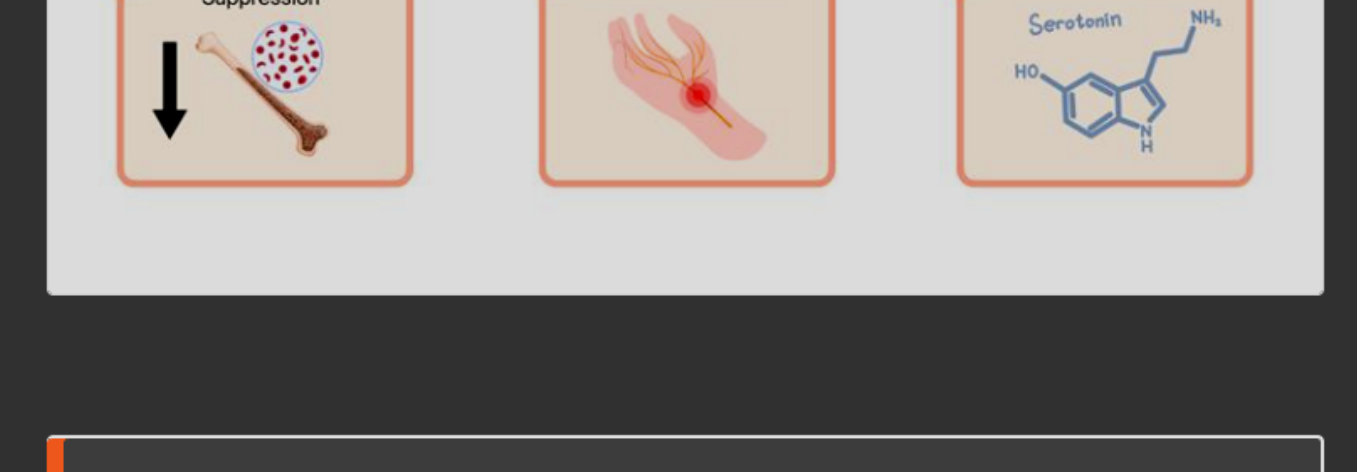
Macrolides and Clindamycin

- Macrolides are **hepatically metabolized**. Erythromycin and Clarithromycin are CYP3A4 substrates and inhibitors. Medications metabolized by CYP3A4 may need to be avoided. Simvastatin and lovastatin should be avoided, while others should be used with caution (eg. Warfarin).
- Macrolides should be use with caution with existing cardiovascular disorders. They can cause **QT prolongation** (a type of arrhythmia) and should be avoided in agents that can increase the risk of QT prolongation.
- Macrolides are contraindicated in patients with cholestatic jaundice or hepatic dysfunction
- Macrolides can cause GI upset (nausea, vomiting, and diarrhea).
- Clindamycin's adverse effects are diarrhea, allergic reaction, and neuromuscular blockade. Pseudomembranous **colitis** from *Clostridioides difficile* overgrowth is often attributed to clindamycin as well.



Linezolid

- Linezolid's adver[illegible]sion, peripheral neuropathy, and serotonin syndrome. Bone suppression and peripheral neuropathy are due to linezolid's mitochondrial protein synthesis inhibition.
- Linezolid are reversible monoamine oxidase (MAO) inhibitors. Monoamine oxidase functions to break down monoamine neurotransmitters such as serotonin, so its inhibition can cause increased levels of serotonin. Co-administration of linezolid with other drugs that increase serotonin levels, such as selective serotonin reuptake inhibitors (SSRIs) and norepinephrine reuptake inhibitors (SNRIs), can result in **serotonin syndrome**.



ACTIVE LEARNING QUESTION

Match the antibiotic to the drug interaction.

Linezolid	
Clarithromycin	✓ Simvastatin
Tetracycline	✓ Antacids
Amikacin	✓ Amphotericin

Explanation

Linezolid could interact with SSRIs to cause serotonin syndrome. Clarithromycin is a CYP3A4 inhibitor; it should be avoided with Lovastatin and Simvastatin. Tetracyclines can chelate with the cations in the antacids. Amikacin use can cause nephrotoxicity. The use of nephrotoxic agents such as Amphotericin could increase risk renal damage.

CLINICAL CORRELATION

Serotonin syndrome is a serious condition caused by an excessive accumulation of serotonin in the body, often due to medications that increase serotonin levels. Symptoms can range from mild, such as shivering and diarrhea, to severe, including muscle rigidity, fever, and seizures. It typically occurs within several hours of taking a new drug or increasing the dose of an existing medication1. If left untreated, sever[illegible]ng.

What are common adverse effects of linezolid?

Streptogramins

- **Arthralgia/myalgias** are very common in patients (>45% of patients).
- An increase in serum creatinine and hyperbilirubinemia (>33% of patients) can also occur

Chloramphenicol

- Chloramphenicol's adverse effects include **blood dyscrasias** (e.g., aplastic anemia, thrombocytopenia, and granulocytopenia). Chloramphenicol's blood cell toxicity is due to inhibition of mitochondrial protein synthesis that causes decre[illegible]which is important in maintaining iron homeostasis.
- **Gray baby syndrome** is specific to chloramphenicol and is attributed to a newborn's (especially in prematurity) lack of UDP-glucuronosyltransferase to metabolize the drug. It is potentially fatal and characteristically presents with hypotension and cyanosis.
- These reactions are very harmful adverse reactions, so it is only recommended in some instances when alternative options are ineffective.

ACTIVE LEARNING QUESTION

Match the PSI to its most common or important adverse effect

Tetracycline	✓ Teeth discoloration
Linezolid	✓ Serotonin syndrome
Amioglycoside	✓ Ototoxicity
Chloramphenicol	✓ Gray baby syndrome
Streptogramin/Dalfopristin	✓ Arthralgia/Myalgia

Explanation

Chelation with the calcium tetracyclines can cause teeth discolorations. Linezolid is a MAO inhibitor, thus can cause an increase in serotonin levels and serotonin syndrome. Aminoglycosides can accumulate in the cochlea and cause ototoxicity. Chloramphenicol cannot be metabolized by newborns, causing Gray baby syndrome. While the cause for arthralgia/myalgia is not known, it is very commonly seen in patients who are on Stretogramins/Dalfopristin.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to KH, what adverse effects will you discuss?

You explain that most people tolerate tetracycline without a problem. "Just so you know what could happen, some people have some nausea, vomiting, or diarrhea. Please let me know if these become a problem for you so we can decide together what to do next. And it is important to take the medicine separate from eating or drinking any dairy products. Those foods can make it impossible for your body to absorb the medicine, so you wouldn't get any benefit from it. Before you start taking it, we need to check a urine pregnancy test. We do this for all women of childbearing age, just to be sure. The medicine would have effects on a growing fetus."

Summary

- **Aminoglycoside**[illegible]icity, with increased risk in patients with impaired renal function and concurrent use of other nephrotoxic drugs.
- **Tetracyclines** can cause gastrointestinal symptoms, teeth discoloration, and photosensitivity; they are contraindicated in children under 8 years and during pregnancy due to potential harm.
- **Macrolides** can cause QT prolongation and are contraindicated in patients with cholestatic jaundice or hepatic dysfunction; they interact with CYP3A4 substrates.
- **Linezolid** can cause bone marrow suppression and serotonin syndrome, especially when combined with SSRIs or SNRIs.
- **Chloramphenicol** can cause blood dyscrasias and gray baby syndrome in newborns due to lack of drug metabolism.
- Adverse reactions to protein synthesis inhibitors range from mild gastrointestinal symptoms to severe effects like nephrotoxicity, ototoxicity, and serotonin syndrome.

Review Questions

1. Which antibiotic is most likely to cause ototoxicity?

A. ● Amikacin
B. ○ Doxycycline
C. ○ Linezolid
D. ○ Quinupristin/Dalfopristin

Hide Explanation

The correct answer is amikacin (A). Amikacin (A) is an aminoglycoside, effective against gram-negative organisms. It can accumulate in the cochlea which can result in ototoxicity. Doxycycline (B) is a tetracycline that is not associated with ototoxicity. Linezolid (C) is most likely to cause bone marrow suppression and also serotonin syndrome. The streptogramins, Quinupristin/Dalfopristin (D) are most likely to cause arthralgia.

2. Which oral antibiotic is most likely to interact with and inhibit CYP3A4?

A. ○ Amikacin
B. ○ Clarithromycin
C. ○ Minocycline
D. ○ Gentamicin

Explanation (requires correct answer)

3. Counselling on photosensitivity and phototoxicity is most relevant when prescribing which class of antibiotics?

A. ○ Aminogly[illegible]
B. ○ Macrolides
C. ○ Streptogramins
D. ○ Tetracyclines

Explanation (requires correct answer)

4. Which antibiotic is contraindicated in a pregnant patient?

A. ○ Azithromycin (oral)
B. ○ Clindamycin (oral)
C. ○ Gentamycin (IV)
D. ○ Streptogramin (IV)

Explanation (requires correct answer)

How would you rate this Brick?

B. ○ Clindamycin (oral)
C. ○ Gentamycin (IV)
D. ○ Streptogramin (IV)

Explanation (requires correct answer)

How would you rate this Brick?

👍 👎

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ **Recognize the common adverse reactions associated with protein synthesis inhibitors.**
02 ☐ **Identify the common drug interactions associated with protein synthesis inhibitors.**

Objective Confidence: 0/2